

NATIONAL UNIVERSITY OF SINGAPORE
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

ACADEMIC YEAR 2025-2026
SEMESTER 1

EE2213: Introduction to AI

QUIZ 1: L1-L4

INSTRUCTIONS TO STUDENTS

1. This assessment consists of three types of questions: True/False Questions, Multiple Choice Questions (only one answer is correct), and Multiple Response Question (allow for multiple correct answers).
2. There are a total of 3 True/False Questions (3 marks each), 9 Multiple Choice Questions (5 marks each), and 8 Multiple Response Questions (7 marks each), for a total of 110 marks. For MRQ (Multiple-Response Questions), marks will be deducted for any incorrect options selected — but your score for each MRQ will never go below zero.
3. Answer all questions.
4. This is an open-book assessment. You may use printed or soft copy materials (e.g., lecture slides, textbooks, notes). However, you are not allowed to search the internet or use any LLM-powered tools (e.g., ChatGPT or other language models).
5. Time allowed: 30 minutes.

Question 1

True or false: The Turing test, proposed by Alan Turing, evaluates whether a machine can solve mathematical problems faster than a human.

- (a) True
- (b) False

Ans: (b) False. The Turing test, proposed by Alan Turing, is designed to assess whether a machine or agent can exhibit behavior that is indistinguishable from that of a human in a conversational setting.

Question 2

True or false: A simple reflex agent selects actions based on the current percept, without any notion of goals.

- (a) True
- (b) False

Ans: (a) True

Question 3

True or false: When designing performance measures for an agent, it's better to base them on the desired behavior of the agent rather than on the outcomes you want the agent to achieve.

- (a) True

(b) False

Ans: (b) False. It is better to design performance measures according to the goals you want the agent to achieve in the environment, rather than prescribing how you think the agent should behave.

Question 4

Which of the following best describes the relationship between Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL)?

- (a) Machine Learning is a broader field that includes AI and Deep Learning.
- (b) Deep Learning and Machine Learning are two separate approaches that are unrelated to AI.
- (c) Artificial Intelligence is a subset of Deep Learning, which itself is a subset of Machine Learning.
- (d) Deep Learning is a subset of Machine Learning, which is a subset of Artificial Intelligence.

Ans: (d)

Question 5

Which of the following statements about rational agents are correct? Select all that apply.

- (a) Rational agents are assumed to know the complete environment dynamics and can predict how the environment will respond to their actions.
- (b) A rational agent chooses actions that are expected to yield the highest performance according to a predefined measure.
- (c) A rational agent that acts rationally will always achieve the true best outcome in every situation.
- (d) Smart home devices, such as automatic lights, can be considered as rational agents.

Ans: (b) (d)

Option (a) is wrong because rational agents are not clairvoyant. They may lack knowledge of the environment dynamics. They don't magically know the future or hidden parts of the environment. Option (c) is also wrong, because a rational agent selects an action that maximizes the expected value of its performance measure, based on the information and knowledge it has at the time. This means a rational agent may not always achieve the *truly optimal* outcome in every situation, since it cannot foresee all future events. Rationality (i.e., agent-optimality) is therefore about making the best possible decision given the agent's knowledge, while actual optimality refers to the actual best outcome in hindsight.

Question 6

In AI, the PEAS framework helps describe an agent's task environment. Which of the following statements about PEAS components are true? Select all that apply.

- (a) PEAS stands for Performance Measure, Environment, Actuators, and Sensors.
- (b) In an autonomous driving agent, steering, brakes, and accelerator are examples of actuators. The Environment includes external elements like roads, pedestrians, and other vehicles.

- (c) Sensors are used to perform physical actions such as turning or stopping.
- (d) The Environment includes everything the agent can interact with or be affected by.

Ans: (a) (b) (d)

Option (c) is wrong. Actuators are used to perform physical actions, not sensors.

Question 7

Which of the following statements about a search problem are true? Select all that apply.

- (a) The transition model defines the successor state resulting from applying an action to a current state.
- (b) The path cost depends only on the starting and ending states, not the sequence of actions taken.
- (c) The goal test determines whether a given state is a goal state.
- (d) Actions are the set of all possible states that the agent can occupy.

Ans: (a) (c)

Option (b) is wrong, as the path cost also depends on the sequence of actions taken. Option (d) is wrong. State space is the set of all possible states of the environment that the agent can occupy.

Question 8

What is the major disadvantage of graph search as opposed to tree search?

- (a) Visual representations of graph search are less appealing but graph search is otherwise the same as tree search
- (b) Graph search is unable to detect repeated states
- (c) There is no disadvantage since graph search is just another name for tree search
- (d) Graph search can require a lot of memory because it keeps track of all possible states

Ans: (d). The key difference between tree search and graph search is that graph search detects and avoids repeated states by maintaining a visited (explored) set. This prevents redundant exploration but requires more memory, since it must store all explored states.

Question 9

Which of the following are reasons to use Depth-First Search (DFS) over Breadth-First Search (BFS)? Select all that apply.

- (a) DFS always finds the shortest path
- (b) DFS guarantees completeness in infinite search spaces
- (c) DFS usually uses less memory than BFS, especially in large or deep search spaces.
- (d) DFS expands all nodes at each level before going deeper
- (e) DFS is good at finding solutions that are many steps away, and supports backtracking, making it more suitable than BFS for complex games like chess and Go

Ans: (c) (e)

Option (a) is wrong. BFS guarantees the shortest path (if all edges have equal cost), whereas

DFS might dive down a long, suboptimal branch first. Option (b) is wrong. In an infinite search space, DFS can get stuck going down one infinite branch and never find a solution (refer to the example in page 10 in Lecture 3). BFS, on the other hand, is complete in finite branching factors (though it usually uses more memory than DFS). Option (d) is also wrong. BFS expands all nodes at each level before going deeper. DFS goes deep along a branch first before backtracking.

Question 10

Which of the following statements about Breadth-First Search (BFS) and Uniform-Cost Search (UCS), also known as Dijkstra's algorithm, are correct? Select all that apply.

- (a) Both BFS and UCS always treat all steps as having equal cost.
- (b) UCS explores paths with the lowest total cost so far, while BFS treats all steps as having equal cost.
- (c) BFS uses actual path costs when deciding which node to expand next, while UCS always expands nodes at the same depth first.
- (d) Both BFS and UCS are complete, assuming positive step costs and finite branching.

Ans: (b) (d).

Option (a) is wrong. UCS (Dijkstra's algorithm) works with arbitrary non-negative step costs and does not assume equal costs. Option (c) is wrong. BFS expands nodes by increasing depth (level order). UCS expands nodes by increasing cumulative path cost.

Question 11

Which of the following about Dijkstra's algorithm is true? Assume the graph search version of Dijkstra's algorithm is used.

- (a) Dijkstra's algorithm can handle graphs with negative edge weights, as long as there are no negative cycles.
- (b) Once a node is removed from the frontier, Dijkstra's algorithm may still revisit it if a shorter path is found later.
- (c) Dijkstra's algorithm assumes that the shortest path to a node is found when it is removed from the frontier, which is only guaranteed to be true if all edge weights are non-negative.
- (d) Dijkstra's algorithm always finds the shortest path, even if some edge weights are negative.

Ans: (c). Dijkstra's algorithm uses a greedy strategy: at each step, it selects the node with the shortest distance from the start. It assumes that once a node is removed from the frontier, its shortest path has been found, but this is only guaranteed when all edge weights are non-negative. Negative edges break this assumption, potentially leading to incorrect results.

Question 12

Which of the following best describes the difference between uninformed and informed search strategies?

- (a) Uninformed search uses heuristics to evaluate nodes, while informed search does not.
- (b) Informed search uses problem-specific knowledge (heuristics) to guide the search

toward the goal, whereas uninformed search does not use any additional information beyond the problem definition.

(c) Uninformed search expands nodes based on their estimated distance to the goal, while informed search explores all nodes equally.

(d) Both informed and uninformed search always expand nodes in alphabetical order to maintain consistency.

Ans: (b)

Question 13

What happens when A* search is used with a heuristic function $h(n) = 0$ for all nodes?

(a) It behaves like Breadth-First Search, ignoring edge costs entirely.

(b) It behaves randomly, since the heuristic provides no guidance.

(c) It behaves like Dijkstra's algorithm, expanding nodes based on the lowest total path from the start node to that node ($g(n)$).

(d) It behaves like Greedy Best-First Search, favoring nodes that appear closer to the goal.

Ans: (c). This is because when $h(n) = 0$, the A* evaluation function $f(n) = g(n) + h(n)$ reduces to $f(n) = g(n)$, which prioritizes nodes with the smallest actual cost from the start, exactly like Dijkstra's algorithm.

Question 14

Which of the following statements about Greedy Best-First Search are correct? Select all that apply.

(a) Greedy Best-First Search is guaranteed to be cost-optimal.

(b) Greedy Best-First Search is guaranteed to be complete.

(c) Greedy Best-First Search uses a heuristic function to estimate the cost to the goal.

(d) Greedy Best-First Search may get stuck in infinite loops or fail to find a solution even if one exists.

Ans: (c) (d).

Question 15

Which of the following statements about admissible and consistent heuristics is true? Assume all heuristics are non-negative.

(a) Every heuristic that is admissible is also consistent.

(b) A consistent heuristic may overestimate the true cost to the goal.

(c) Every consistent heuristic is also admissible.

(d) Admissibility is a stricter condition than consistency.

Ans: (c). An admissible heuristic never overestimates the true cost to the goal, while a consistent heuristic also satisfies the triangle inequality. Therefore, every consistent heuristic is admissible, but not every admissible heuristic is consistent.

Question 16

Which of the following statements about informed search algorithms are true? Select all that apply.

- (a) The efficiency (in terms of time and number of node expansions) of informed search algorithms is influenced by the quality and informativeness of the heuristic function.
- (b) A better heuristic can reduce the number of nodes explored during the search.
- (c) Informed search algorithms always find the optimal solution regardless of the heuristic quality.
- (d) If $h(n)$ is a consistent heuristic and all step costs are positive, then the total estimated cost $f(n) = g(n) + h(n)$ never decreases along any path, where:
 - $g(n)$ is the cost to reach node n from the start node, and
 - $h(n)$ is the estimated cost from node n to the goal.

Ans: (a) (b) (d). Informed search algorithms rely heavily on the heuristic: a more accurate heuristic leads to fewer node expansions and greater efficiency, making (a) and (b) true. They do not always guarantee optimality unless the heuristic is admissible/consistent, so (c) is false, while (d) is true because with a consistent heuristic and positive costs, $f(n)$ is non-decreasing along any path.

Question 17

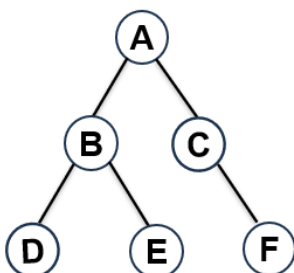
Which of the following statements about A* search are true? Select all that apply.

- (a) A* search relies on both the path cost so far and the heuristic estimate to the goal.
- (b) A* search is complete and will return a solution if one exists, provided the branching factor is finite and step costs are non-negative.
- (c) A* search with the tree search strategy is optimal if the heuristic is admissible and all step costs are non-negative.
- (d) A* search with the graph search strategy is optimal if the heuristic is consistent and all step costs are non-negative.

Ans: (a) (b) (c) (d). *Reminder: This question focuses on A* search, a key algorithm in the Search topic—make sure you understand its definition and properties.*

Question 18

Consider the graph below. Starting from node A, and assuming neighbors are visited in alphabetical order, which of the following shows the correct order in which nodes are visited using Breadth-First Search (BFS) and Depth-First Search (DFS), respectively? Assume a graph search strategy for both BFS and DFS, where nodes are not revisited once processed.



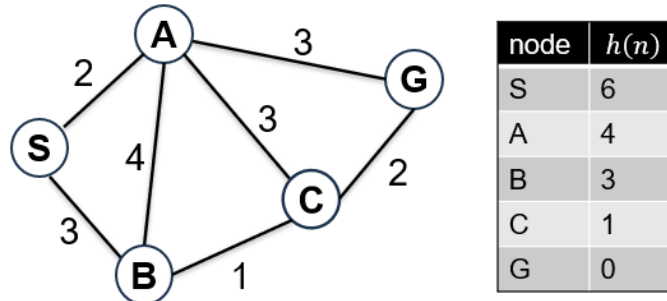
- (a) BFS: A, B, C, D, E, F
DFS: A, B, D, E, C, F
- (b) BFS: A, B, C, F, D, E
DFS: A, C, F, B, E, D
- (c) BFS: A, B, C, D, F, E
DFS: A, B, E, D, C, F
- (d) BFS: A, B, C, D, E, F
DFS: A, B, D, E, F, C

Ans: (a). To solve this, start at node A and expand nodes level by level for BFS, always visiting neighbors in alphabetical order and avoiding revisits (because we are using graph search strategy). For DFS, follow one branch as deep as possible before backtracking, again choosing neighbors alphabetically and not revisiting nodes (graph search strategy). By applying these rules, you can trace the visit sequences to identify the correct option.

Question 19

Please perform greedy best-first search on the graph below, using a graph search strategy (i.e., nodes are not revisited once expanded). The numbers on the edges represent the step cost between nodes, and the heuristic values (estimated cost to the goal G) for each node are shown in the table on the right side of the graph.

Starting from node S, which of the following is a possible total cost of the path found by greedy best-first search to reach the goal node G?

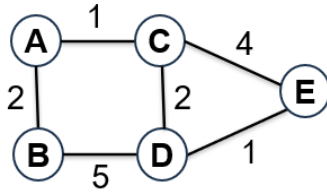


- (a) 5
- (b) 6
- (c) 7
- (d) 8
- (e) 9

Ans: (b). Visiting sequence (based solely on heuristic values): $S \rightarrow B \rightarrow C \rightarrow G$

Question 20

Consider the graph below. Starting from node A, which of the following shows a possible order in which nodes are expanded (i.e., popped from the priority queue) by Dijkstra's algorithm, assuming a graph search strategy where nodes are not revisited once processed? The numbers on the edges represent the cost of each edge.



- (a) A, C, D, E, B
- (b) A, B, C, D, E
- (c) A, C, E, D, B
- (d) A, C, B, D, E

Ans: (d).

Solving steps:

1. Initialize: Start at node A. Set the cost to reach A as 0 and all other nodes as infinity. Place A in the priority queue.
2. Expand the lowest-cost node: Pop the node with the smallest total cost ($g(n)$) from the priority queue. Initially, this is A.
3. Update neighbors: For each neighbor of the current node, calculate the total cost to reach it through the current node. If this cost is smaller than the previously recorded cost, update it and add the neighbor to the priority queue.
4. Repeat: Continue popping the lowest-cost node from the priority queue and updating its neighbors, without revisiting nodes already processed (because we are using graph search strategy).
5. Trace the order: As you expand nodes in order of increasing path cost, record the sequence of nodes popped from the queue. This sequence is what the question asks for.